

- Project overview:

part 1: test

part 2: make the reinforcement learning, the goal is to make the total acc up to at least 70%

part 3:

The goal to the project is to find the kaggle in a map. There are two robot with A* algorithm to find the kaggle together.

The A* heuristic is manhattan distance.

Two Bot will regard each other as a block.

Bot A's target is kaggle, while Bot B's target is Bot A initially

If the Bot B's distance to kaggle is less than 5 step, Bot B's target becomes kaggle.

- How to run:

part 2:

python ./frozen_lake.py

part 3:

main.py

You can change the variable ****RENDER_FLAG**** to determine if you want the experiment mode or render mode.

ender mode: Only execute twice, first time will demonstrate how two bot work together, second time will demonstrate only Bot A working as comparison.

Experiment mode: The mode will demonstrate how it works if there are 1000 kaggle(default, you can change it).

Map size: You can adjust the variable ****GRID_R**** and ****GRID_C**** to whatever you want.

MAX_MISSION: This variable means how many kaggles are there, but you can change it in ****Render Mode****.

MAX_STEPS_PER_MISSION: This variable is the limit step to a mission.

execute

python pygame

python main.py

- Dependencies:

part 2:

pickle numpy matplotlib gymnasium

part 3:

pygame

Graph Render and environment simulate

sprites

Render the picture of bot, floor and kaggle

- Contribution list (who did what):

student id	name	contibution	content
B123040030	劉兆涵	40%	finish pat2
B123040031	許耕瑜	30%	finish part3
B123040010	王紹庭	30%	finish ppt